High Integrity GNSS Navigation and Safe Separation Distance to Support Local-Area UAV Networks

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ABSTRACT

A local-area Unmanned Aerial Vehicle (UAVs) network can provide local-area differential corrections and optimized guidance, including the maintenance of safe separation distances, for precise and safe operation of the UAVs. In networked UAV operations, only the uncorrelated component of position error between two UAVs in the same network contributes to potential separation violations, because "in-network" UAVs share the same source of guidance and differential corrections. Therefore, models of uncorrelated errors need to be defined to establish safe separation distances between UAVs.

This paper develops a methodology to estimate safe separation distance for UAVs which share the same source of guidance and local-area differential corrections. Using this methodology, ionospheric and tropospheric models for Navigation System Error (NSE) are developed theoretically. The airborne multipath error and Flight Technical Error (FTE) components which depend on hardware, environment, and operational conditions are also determined though UAV flight experiments. The standard deviation of FTE was estimated in both straight and curved flight segments. The results show that a lateral FTE error of 0.78 meters in curved segments is much greater than lateral FTE in straight segments and vertical FTE in both segments (all about 0.4 meters). This is due to the momentum of the UAV when it is taking turns and the limited controller response time of the UAV.

The flight experiments also show that UAV multipath errors are reasonably well-bounded by the standard airborne multipath model developed for the Ground Based Augmentation System (GBAS). From the estimated error models, simulations of safe separation distances were conducted under the "in-network" UAVs scenario. Simulation results for a 24-satellite GPS constellation and the probabilities of safe separation suggested in prior work show that vertical separations between UAVs in the same network vary from about 3.0 – 6.5 meters, while horizontal separations vary between 2.2 and 3.5 meters. These values change with time according to the visible GPS satellite geometry and are known to the controller in real time.

1.0. INTRODUCTION

Recent technological improvements and increased military utilization have proven the operational viability of unmanned aerial vehicles (UAVs) and made them attractive for a wide range of potential civil applications. In particular, the capability and low cost of small UAVs have led to a dramatic growth in potential commercial applications over the past few years as well as generating major interest in the media. While most current UAVs are remotely piloted by humans, future UAVs will operate autonomously and coordinate their activities in large groups. Autonomous UAVs networks guided by central controllers will safely and routinely perform various missions such as surveillance, reconnaissance, and data collection [1].

Prior work [1, 2] has proposed a concept of UAV network operations that uses differential corrections from Local-Area Differential GNSS (LADGNSS) to provide increased accuracy, safety, and reliable navigation to UAVs within 5 to 100 km of the controller station. LADGNSS supports low cost for commercial applications and high integrity to allow UAVs to operate in close proximity to each other while minimizing collision risk. Local-area UAV networks can perform their tasks safely as long as UAV locations within the network are

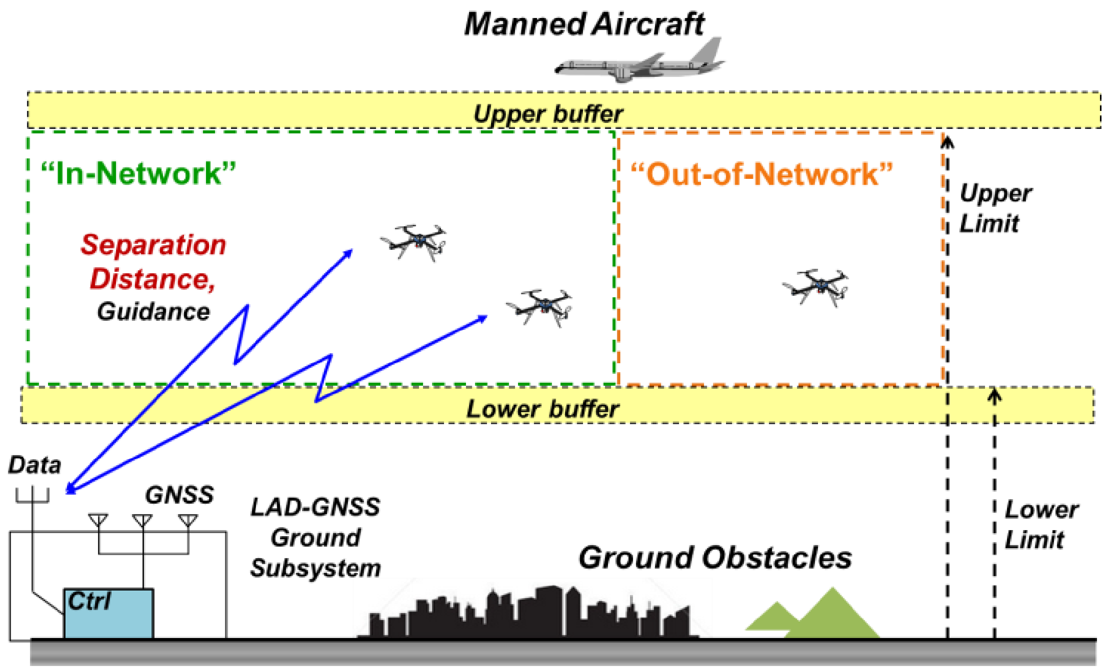

Figure 1. Concept of Local-Area UAV Networks

monitored by the controller to insure that safe separation distances are maintained from other vehicles and ground obstructions. The controller, including a LADGNSS reference station, receives GNSS signals and positions of UAVs, estimates errors and separation distances, and broadcasts differential corrections and optimized guidance for precise and safe operation.

This paper presents a methodology to estimate the minimum separation distance that should be maintained to prevent collisions as part of autonomous UAV guidance. Separation distance can be derived by modeling two sources of position uncertainty that could potentially lead to collisions: (i) Navigation System Error (NSE), and (ii) Flight Technical Error (FTE). This paper models these errors both theoretically and experimentally. Ionospheric and tropospheric models for NSE are developed theoretically. Through UAV flight experiments, we also determined models for airborne NSE and FTE in different flight modes. In this paper, Section 2.0 illustrates the concept of local area UAV networks and how safe separation distances between UAVs are defined. Section 3.0 explains the methodology for estimating safe separation distance, including the details of how to model NSE and FTE. Section 4.0 shows the UAV and network test configuration used for flight experiments. Section 5.0 shows the results of UAV flight tests and simulation of safe separation distance between UAVs in the same network. Section 6.0 concludes the paper.

2.0. LOCAL AREA UAV NETWORK OVERVIEW

Figure 1 shows the concept of local-area UAV networks proposed in [1,2]. The control station, including a LADGNSS reference station, is the source of local-area differential corrections and integrity information as well as optimized guidance driven based on safe separation distances. This allows the UAVs to operate in close proximity to each other with the minimal risk of collision. The maximum operational range of a single network can be horizontally limited by the coverage of data-link rather than the gradual accuracy degradation of LADGNSS differential corrections. For the vertical boundaries of operations, the UAVs fly in the space between the upper and lower limits, which are defined by the minimum distances below manned aircrafts and above ground obstacles respectively.

In Figure 1, "in-network" UAVs refer to the UAVs within the same network which share the same source of guidance and LADGNSS differential corrections. The "In- network" UAVs have both vertical and horizontal minimum separations between themselves for safe operations. "Out-of-network" UAVs represent all the other UAVs. They may share the use of GNSS, but do not share guidance or differential corrections. As they are assumed to fly in the same general airspace allocated to the UAVs, separations between "in-network" and "out-of-network" UAVs are also defined to avoid collisions with each other.

As shown in Figure 2, the separation distances are divided into mainly two error sources that could potentially lead to collisions. The first source is NSE, or error in the estimated position (compared with true position). This is caused by residual pseudorange errors which cannot be completely eliminated by applying LADGNSS differential corrections. The residual errors consist of four error components: residual tropospheric error, residual ionospheric error, ground facility error, and airborne error, which is the Root-Sum-Square (RSS) of airborne multipath error and receiver noise. As for the second source, FTE, error in following the path defined by the guidance source, depends on the controller's performance and external disturbances such as wind. In Figure 2, "Target" is an obstacle to be avoided and can represent another vehicle, including "in-network" and "out-of-network" UAVs.

In this paper, we focus on estimating separation distances between "in-network" UAVs. The target in Figure 2 is now other UAV in the same network. As LADGNSS navigation errors for nearby "in-network" UAVs will be very highly correlated, only uncorrelated error components contribute to the potential separation violations. To compute safe separation distances, the uncorrelated errors among UAVs should be accurately modeled.

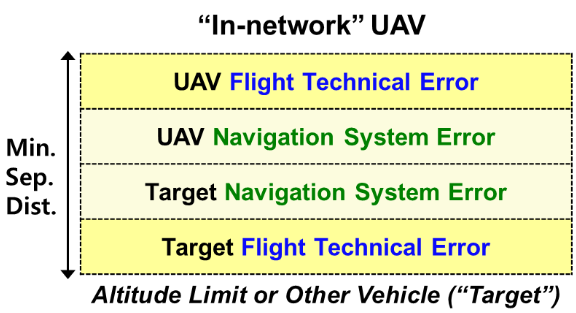

Figure 2. Elements of Required Separation Distance

3.0. METHODOLOGY FOR ESTIMATING SAFE SEPARATION DISTANCE

In a local-area UAV network which uses LADGNSS as a means of navigation, residual navigation errors still remain even after applying differential corrections. However, since correlated errors affect in-network UAVs in the same manner, only the uncorrelated error components contribute to the potential separation violations. The uncorrelated residual errors were theoretically modeled for troposphere and ionosphere, respectively. We also determined the models for error components which depend on hardware, environmental and operation conditions experimentally. These errors include airborne multipath, receiver noise and multipath at the ground reference station, and flight technical error. The ground facility errors included in the position information of UAVs within the same network are common since UAVs shares the same ground station. Thus, the ground facility errors are not included in the computation of separation distances among in-network UAVs.

3.1. RESIDUAL TROPOSPHERIC ERROR MODELING

As each UAV flies at a different altitude relative to the ground reference stations, a differential tropospheric delay correction is required. For Ground Based Augmentation System (GBAS) applications, the vertical tropospheric correction in meters for a user at an altitude of Δh meters above the reference station can be given by

$$TC_{vert} = 10^{-6} N_R h_0 (1 - e^{-\Delta h / h_0}) \tag{1}$$

where

N_R= refractivity index transmitted by ground subsystem [3].

h_0 = tropospheric scale height transmitted by ground subsystem (in meters) [3].

For a satellite at an elevation angle θ, the slant (i.e., along the actual path between satellite and receiver) tropospheric correction is defined as follows:

$$TC = \frac{N_R h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (1 - e^{-\Delta h / h_0}) \tag{2}$$

Using the refractivity uncertainty, σ_N , transmitted by ground subsystem [3], the residual tropospheric error bound for GBAS users is defined by

$$\sigma_{tropo} = \frac{\sigma_N h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (1 - e^{-\Delta h / h_0}) \tag{3}$$

The uncorrelated residual tropospheric error between two in-network UAVs are computed by differentiating tropospheric corrections for UAV_a at Δh_a meters and UAV_b at Δh_b meters above the reference station. Since correlated errors between two UAVs are canceled out, the uncorrelated residual tropospheric error bound between UAVs is computed as

$$\begin{aligned} \sigma_{tropo,betweenUAVs} &= \sigma_{tropo}(UAV_b) - \sigma_{tropo}(UAV_a) \\ &= \frac{\sigma_N h_0 \times 10^{-6}}{\sqrt{0.002 + \sin^2(\theta)}} (e^{-\Delta h_b / h_0} - e^{-\Delta h_a / h_0}) \end{aligned} \tag{4}$$

3.2. RESIDUAL IONOSPHERIC ERROR MODELING

Ionospheric temporal and spatial decorrelation can lead to residual range errors to LADGNSS users. The temporal decorrelation refers to the change of ionospheric delay over time at the same location in the ionosphere. If users apply the same carrier-smoothing time constant with a reference station, the temporal decorrelation does not lead to steady state residual range error [4]. On condition that carrier-smoothing time constants are matched between reference station and in-network UAVs, the spatial decorrelation, the change of delay from point to point in the ionosphere at a given time, is only a matter of interest.

The derivation of ionospheric spatial decorrelation error model between UAVs is an extension of the original work, which derived a simplified model (σ_{iono}) of residual range error due to spatial decorrelation for GBAS users. The original model was for ionospheric decorrelation errors between a fixed reference location and an approaching aircraft. In this paper, we develop a generalized model of ionospheric decorrelation error experienced between two moving UAVs. The differential carrier-smoothed ionosphere delay between UAV_a and UAV_b at epoch k ($\Delta\hat{I}_k$) can be written as [5]

$$\Delta\hat{I}_k = \frac{\partial I_r}{\partial x_r} x_r - 2(\tau - T_s)\frac{\partial I_b}{\partial x_b} v_{b,k-1} + 2(\tau - T_s)\frac{\partial I_a}{\partial x_a} v_{a,k-1} \quad (5)$$

where x_r is the range distance (in meters) between UAV_a and UAV_b ; $\partial I_r / \partial x_r$ is the spatial ionospheric gradient between UAVs; τ is the carrier smoothing time constant (in seconds); T_s is the sampling time (in seconds); $\partial I_a / \partial x_a$ and $\partial I_b / \partial x_b$ are the spatial ionospheric gradients in the direction of motion of UAV_a and UAV_b respectively; and v_a and v_a are the velocities of the UAV_a and UAV_b respectively. As shown in Equation (5), the differential ionospheric delay consists of error components caused by the spatial ionospheric decorrelation between the UAVs, which are expressed as $\partial I_r / \partial x_r \cdot x_r$, and the smoothing filter effects.

To derive the residual ionospheric error model, two conditions are assumed. First, we assume a spatially linear ionospheric decorrelation with a gradient of G as shown in Figure 3. ψ represents an angle between semi-infinite ionospheric structure with a gradient of G and line of sight between UAVs. θ_a and θ_b are angles between the line of sight between UAVs and velocity vectors of UAV_a and UAV_b respectively. Given this assumption, $\partial I_r / \partial x_r$, $\partial I_a / \partial x_a$ and $\partial I_b / \partial x_b$ can be obtained as

$$\frac{\partial I_r}{\partial x_r} = G\cos\psi \quad (6)$$

$$\frac{\partial I_a}{\partial x_a} = G\cos(\psi + \theta_a) = G\left(\cos\psi\cos\theta_a - \sin\psi\sin\theta_a\right) \quad (7)$$

$$\frac{\partial I_b}{\partial x_b} = G\cos(\psi + \theta_b) = G\left(\cos\psi\cos\theta_b - \sin\psi\sin\theta_b\right) \quad (8)$$

Therefore, the differential carrier-smoothed ionospheric delay can be formulated as

$$\begin{aligned}\Delta\hat{I}_k = {} & \\ G\Big[& \cos\psi\left(x_r - 2(\tau - T_s)v_{b,k-1}\cos\theta_b + 2(\tau - T_s)v_{a,k-1}\cos\theta_a\right) \\ & + \sin\psi\left(2(\tau - T_s)v_{b,k-1}\sin\theta_b - 2(\tau - T_s)v_{a,k-1}\sin\theta_a\right)\Big]\end{aligned} \quad (10)$$

Using the trigonometric function, the differential delay can be simply expressed as

$$\begin{aligned}\Delta\hat{I}_k &= A\cos\psi + B\sin\psi = \sqrt{A^2 + B^2}\,\sin(\psi + \varphi), \\ \sin\varphi &= \frac{A}{\sqrt{A^2 + B^2}}\end{aligned} \quad (11)$$

To conservatively bound the errors, secondly we assume that the ionospheric slope is located in the direction that maximizes the differential delay between UAVs. This is done by determining ψ which makes $\sin(\psi + \varphi)$ in Equation (11) to be one. Then, using one-sigma value for nominal vertical ionospheric spatial gradients (σ_{vig}) [3], the residual ionospheric error bound between UAVs is derived as

$$\begin{aligned}&\sigma_{iono,betweenUAVs} = \\ &F_{pp}\sigma_{vig}\sqrt{\begin{aligned}&\left(x_r - 2(\tau - T_s)v_{b,k-1}\cos\theta_b + 2(\tau - T_s)v_{a,k-1}\cos\theta_a\right)^2 \\ &+\left(2(\tau - T_s)v_{b,k-1}\sin\theta_b - 2(\tau - T_s)v_{a,k-1}\sin\theta_a\right)^2\end{aligned}}\end{aligned} \quad (12)$$

where F_{pp} is the vertical-to-slant obliquity factor for a given satellite.

$$F_{pp} = \left[1 - \left(\frac{R_e\cos\theta}{R_e + h_I}\right)^2\right]^{-\frac{1}{2}} \quad (13)$$

where
R_e = radius of the Earth = 6378.1363 km.
h_I = ionospheric shell height = 350 km.
θ = elevation angle of satellite.

If the proposed model is applied to the scenario of GBAS users, ionospheric decorrelation between a fixed reference location and an approaching aircraft is modeled by setting v_a to be zero, v_b to be v_{air}, θ_a to be 0°, and θ_b to be 180°. We obtain the original GBAS ionospheric error model [3] as following:

$$\sigma_{iono} = F_{pp} \times \sigma_{vig} \times \left(x_r + 2 \times \left(\tau - T_s \right) \times v_{air} \right) \quad (14)$$

This case study verifies that the proposed ionospheric model is a generalized version of ionospheric decorrelation error model encompassing the original model.

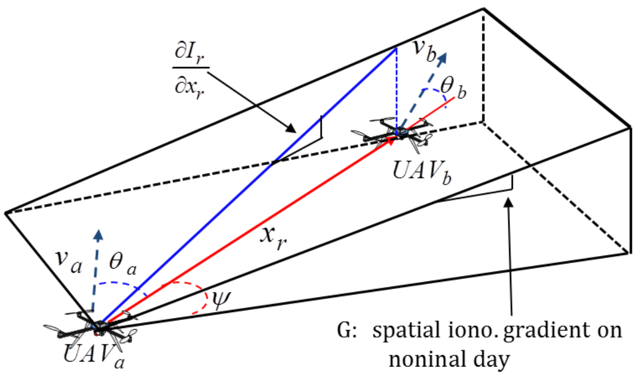

Figure 3. Illustration of Differential Ionospheric Delay between UAV_a and UAV_b

3.3. AIRBORNE MULTIPATH MODELING

As UAVs fly relatively close to ground objects, significant airborne multipath can be presented from ground reflections. The airborne multipath errors also are influenced by the entire airborne hardware and software implementation including flight platforms, antennas, receivers, and code-carrier smoothing filters. Therefore, the airborne error is modeled using empirical data obtained from UAV flight experiments. To estimate multipath errors, dual frequency GPS data were collected at the UAV platforms. The overall data analysis process is illustrated in Figure 4.

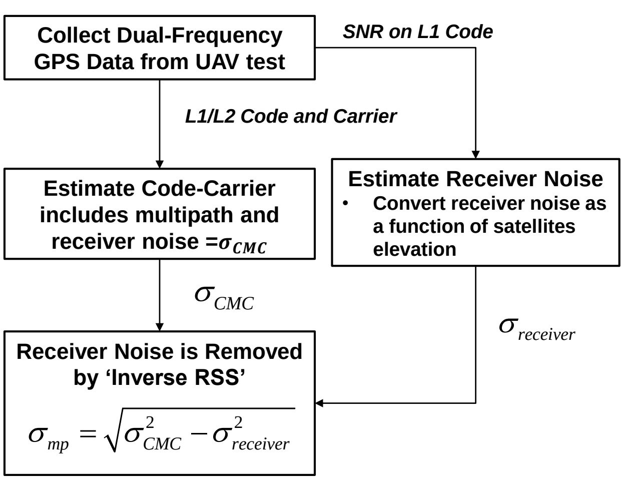

Figure 4. Data Analysis Process of Multipath Estimation

The dual frequency GPS data were analyzed by differentiating L1 code and carrier phase observables for each satellite as shown in Equation (15).

$$\rho_{L1} - \phi_{L1} = 2I + M_{L1} - m_{L1} - N_{L1} + \varepsilon_\rho + \varepsilon_\phi \quad (15)$$

Differentiating the code and carrier phase observables leaves code and carrier multipath errors (M_{L1} and m_{L1}), receiver noise errors on code and carrier measurements (ε_ρ and ε_ϕ), doubled ionospheric errors due to code-carrier divergence (I), and an integer ambiguity (N_{L1}). In this equation, the multipath and receiver noise errors on carrier-phase observables are small enough to be negligible and the integer ambiguity can be considered as a removable bias as long as no cycle slips exist. After estimating and correcting for the ionospheric delay by using the dual frequency carrier phase observables, we obtain the estimates of multipath and receiver noise errors on L1 code-phase observables for each satellite as shown in Equation (17).

$$I = \frac{\phi_{L1} - \phi_{L2}}{\gamma - 1} \quad (16)$$

$$\rho_{L1} - \phi_{L1} - 2\frac{\phi_{L1} - \phi_{L2}}{\gamma - 1} = M_{L1} + \varepsilon_\rho \quad (17)$$

If the multipath and receiver noise are independent and all other error sources are effectively removed from the Code-Minus-Carrier (CMC) observations, then the thermal noise is removed by 'inverse RSS':

$$\sigma_{mp} = \sqrt{\sigma_{CMC}{}^2 - \sigma_{noise}{}^2} \quad (18)$$

The standard deviation of receiver noise was computed using the theoretical model as shown in Equation (19) and it was subtracted from CMC estimates [4]. The theoretical receiver noise model we used is expressed as

$$\sigma_{noise} = \lambda_c \sqrt{\frac{d}{8C/N_0\tau}} \quad (19)$$

where λ_c is the code chip wavelength (in meters), d is the correlate spacing, C/N_0 is the carrier to noise value in units of ratio-Hz, and τ is the smoothing time constant.

3.4. FLIGHT TECHNICAL ERROR MODELING

In order to estimate FTE, firstly, it is required to set up path-following guidance for UAVs. In this research, the flight path is designed to cover areas of interest considering practical applications of UAVs such as agriculture, surveying, mapping, and reconnaissance. Figure 5 shows the designed flight path that consists of three straight sections and two semicircular sections. The ground altitude of each waypoint is fixed as 10 meters. UAV flight tests were conducted along the path, and the flight trajectory information was logged at 10 hertz during the tests.

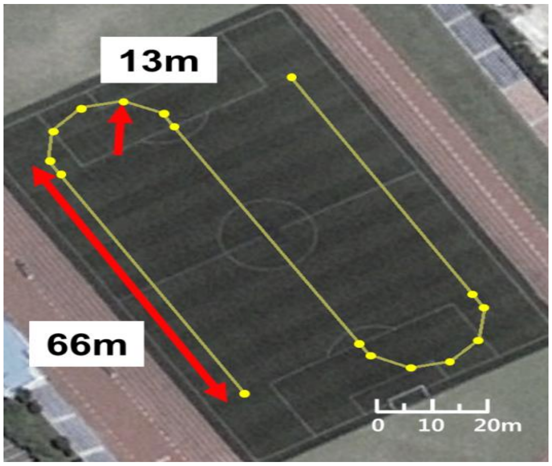

Figure 5. Flight Path of UAV Flight Experiments

The lateral and vertical FTE are calculated by subtracting the designed path position from the logged GPS position in each segment. Then the standard deviation of FTE is computed based on an assumption that samples are independent. To meet this assumption, we resampled data points along the flight path with a constant distance interval in space shown in Figure 6 [6]. This interval is determined by multiple hypothesis tests which were performed until the correlation of sample points become smaller than a threshold. First, in order to evaluate the significance of autocorrelation at the k[th] lag for N samples, the autocorrelation function is defined as follows:

$$\gamma_k = \frac{\sum_{i=1}^{N-k}(x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^{N}(x_i - \bar{x})^2} \tag{20}$$

where x_i is the i[th] data sample and $\bar{x}$ is the mean of data samples. The significance of autocorrelation at the k[th] lag is tested to determine the distance interval between samples. The null hypothesis that the k[th] lag autocorrelation value is statistically zero can be expressed as:

$$H_0 : \rho_k = 0 \tag{21}$$

The hypothesis is evaluated with the critical values for the type I error level α. The type I error level α is the probability of event that the null hypothesis is rejected when it is true. The critical value is defined as follows:

$$\gamma_{\alpha,k} = N^{-1}(\alpha,0,1)\sqrt{\frac{1}{n}\left(1+2\sum_{i=1}^{k}\gamma_i^2\right)} \tag{22}$$

where $k = 1,\cdots,\text{int}(N/4)$ and $N^{-1}(\alpha,0,1)$ is an inverse of standard normal cumulative distribution function at α. The null hypothesis at the k[th] lag is rejected if $\gamma_k > \gamma_{k,\alpha}$.

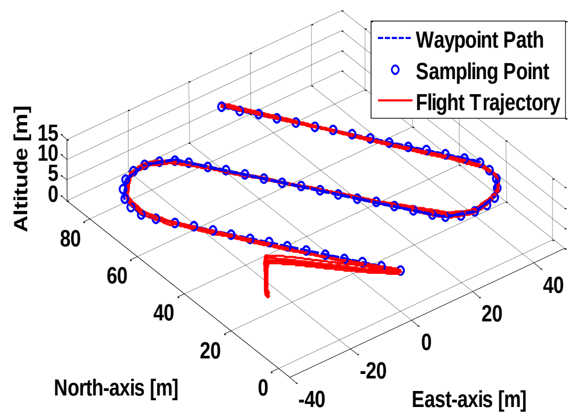

Figure 6. Waypoint Path (Defined Guidance path), Flight Trajectory (Estimated Position) and Sampling Points with Constant Interval

3.5. SAFETY RISK ASSESSMENT

The proposed error models were used to calculate the minimum separation distances that satisfy the probability of collision obtained from the risk-assessment methodology described in [2, 7, 8, 9]. The *K*-factor derived from the risk assessment methodology is multiplied to the error models in order to estimate the separation distances as shown in Equation (23).

$$\textit{Safe Separation Distance} = K \times \sqrt{\sigma_{NSE}^2 + \sigma_{FTE}^2} \tag{23}$$

$$\sigma_{NSE,lat} = \sqrt{\sum_{i=1}^{N} s_{lat,i}^2(\sigma_{tropo,i}^2 + \sigma_{iono,i}^2 + \sigma_{air,i}^2)}$$

$$\sigma_{NSE,vert} = \sqrt{\sum_{i=1}^{N} s_{vert,i}^2(\sigma_{tropo,i}^2 + \sigma_{iono,i}^2 + \sigma_{air,i}^2)} \tag{24}$$

$$\sigma_{air,i} = \sqrt{\sigma_{mp,i}^2 + \sigma_{noise,i}^2}$$

where

N= number of satellites.

$s_{lat,i}$ = projection of the lateral component for the i^{th} ranging source.

$s_{lat,i}$ = projection of the vertical component for the i^{th} ranging source.

The probability that an "in-network" UAV collides with other vehicles and thereby causes harm to human is the key to determining how much error must be allowed for. A collision of UAV in the air has the possibility of falling to the ground. The wreckage dropped may lead to harm to nearby people. Therefore, the probability of harm to humans (state "HH"), given a collision (state "CO") resulting from violation of separation standards (state "SV") can be defined as

$$\Pr(HH \mid SV) = \Pr(HH \mid CO) \times \Pr(CO \mid SV) \quad (25)$$

Table 1. Collision and Hazard Probabilities Given Probability of Separation Violations

Scenario	Pr(CO|SV) (mean)	Pr(HH|CO) (mean)	Pr(HH|SV) (mean)	Pr(HH|SV) (worst case)
In-network UAV	10^{-4}	10^{-5}	10^{-9}	10^{-5}

The Table 1 that shows the probabilities of collisions leading to harm to humans given violation of the minimum separation was proposed in [2]. The "worst case" probability assumed that the probability of collision given a violation of separation is 1.0. Although the mean probability of collision given a violation of separation is very small, the traditional means of assuring safety under anomalous conditions may require the assumption of "worst-case" conditions rather than average ones. Using this risk model with the "worst-case" condition, the probability of the violation of separation is determined when the probability of harm to humans is given. We can borrow 10^{-9} that is the smallest probability required by the safety standard in the FAA hazard risk index [7]. If 10^{-9} is the highest allowed risk of harm to humans that can be tolerated from the violation of UAV separation, given in Equation (26), the probability of separation violation is determined as 10^{-4}.

$$\Pr(HH \mid SV) \times \Pr(SV) \le 10^{-9} \quad (26)$$

Therefore the Gaussian K-factor of 3.89 is determined to represent a double-sided tail probability of 10^{-4}.

4.0. FLIGHT TEST CONFIGURATION

As shown in Figure 7, the LADGNSS test-bed was installed on the KAIST mechanical engineering building, and UAV flight tests were conducted at the KAIST stadium. These two sites are located about 950 meters apart. The differential corrections generated by the LADGNSS Test-bed are available to users anywhere on the campus. The ground facility of LADGNSS requires multiple reference receivers to generate corrections and to detect receiver failures. Figure 8 shows three antennas (connected to three NovAtel ProPak-V3 receivers) mounted on the rooftop of the mechanical engineering building. The antennas are separated out to reduce the correlations between multipath errors of each receiver. However, unlike GBAS installation at airports, they do not spread over a few hundreds of meters due to siting constraints. This setup is somewhat similar to the potential commercial applications for which the spread of antennas is limited.

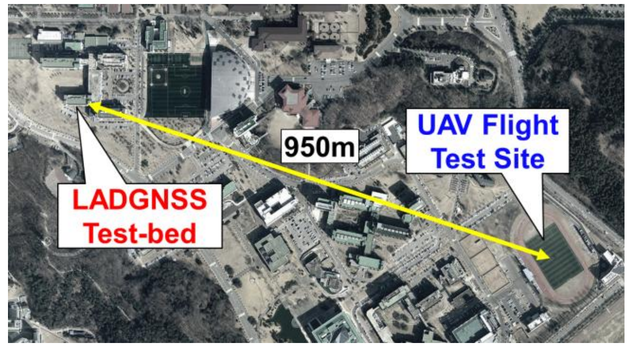

Figure 7. LADGNSS Test-bed and UAV Flight Test Site at KAIST

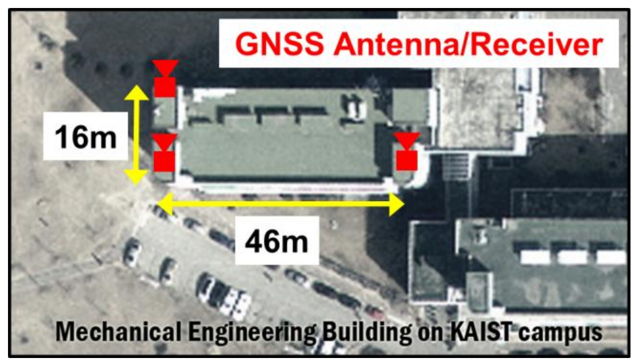

Figure 8. Configuration of LADGNSS Test-bed at KAIST

Multi-copter UAV platform is selected for flight tests, because it ease to fly and control in constricted locations and has very high levels of maneuverability. Figure 9 shows the multi-copter platform used for the flight tests. It is an octo-copter with the size of 110 cm in diameter, 35 cm in height, and 5.26 kg in weight. This vehicle is capable of approximately 10 minutes continuous flight time with two of 22.2V-5000mAh Lithium-polymer batteries. We loaded flight equipment onto the octo-copter to receive LADGNSS differential corrections, estimate precise positions, and fly autonomously. This equipment includes a modem, a NovAtel OEM-V3 receiver, a NovAtel antenna, and Ardupilot Mega (APM) 2.6 controller which uses PID feedback control base.

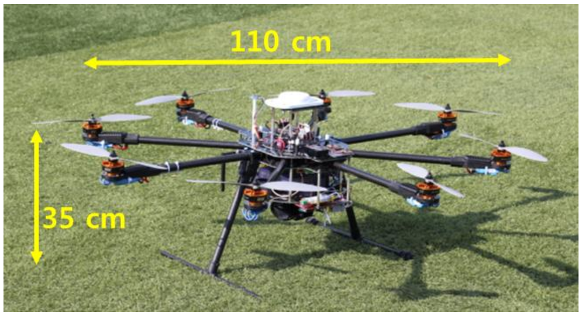

Figure 9. Multi-copter UAV Platform for Flight Test

5.0. RESULT

5.1. AIRBORNE MULTIPATH MODELING

Throughout the UAV flight tests, data was collected from the NovAtel receiver using a 0.1-chip C/A-code correlator spacing. The UAV multipath errors were estimated by subtracting integer ambiguity, ionospheric delay, and theoretical receiver noise from CMC observations as described in Section 3.3. A total of 49 flight tests were conducted, which corresponds to a total flight time of 4 hours 3 minutes. In these flights, the UAVs hovered at the flight altitude of 40 m, which is lower than that of manned aircraft, but well within the flight region of UAVs. Hovering this close to the ground, the measured errors most likely include multipath errors from ground reflections (as well as reflections off UAV surfaces). The multipath error estimates are sorted onto 5° elevation bins, and the standard deviation is computed for each bin. Figure 10 shows the standard deviation of multipath errors for the data from all UAV flight tests compared to the standard airborne multipath model proposed in [3], which is given by:

$$\sigma_{Std.multipath}(\theta) = 0.13 + 0.53e^{(-\theta/10^{\circ})} \qquad (27)$$

The UAV flight test data (blue curve in Figure 10) are reasonably well-bounded by the standard multipath model shown as the red curve in Figure 10. Therefore, it is reasonable to use the standard multipath error model for bounding the UAV multipath errors for this altitude and environment. Future work will include tests at lower altitudes and different ground environments. The degree of multipath correlation between multiple "in-network" UAVs under various flight conditions will be also investigated.

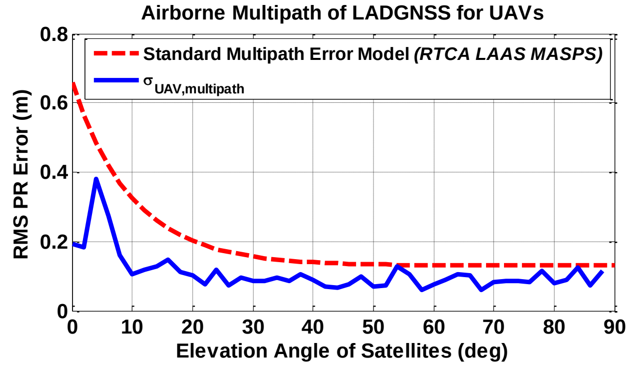

Figure 10. Observed UAV Multipath Compared with Standard Model

5.2. FLIGHT TECHNICAL ERROR MODELING

A total of 21 UAV path-following flights were conducted, and 51,236 FTE samples were collected. Following the resampling process noted in Section 3.4, the optimal sampling distance was determined to be 5.15 meters when the Type I error level α is 99.5%. As a result of the resampling, only 1,239 independent FTE samples are left. FTE error data are categorized into FTE in straight segments and FTE in curved segments. Figure 11 shows that lateral FTE RMS error in curved segments is much greater than the other FTE errors, which are all about the same. This is because the lateral FTE in curved segment is directly influenced by the momentum of the UAV. In addition, the UAV should control its attitude on the horizontal plane more frequently when it makes turn around than when it just go straight, so it can frequently incur the controller errors. On the other hand, the difference between the vertical FTEs of straight and curved segments is very small. This is because there is no variation in vertical guidance. These factors are highly dependent on the flight controller itself.

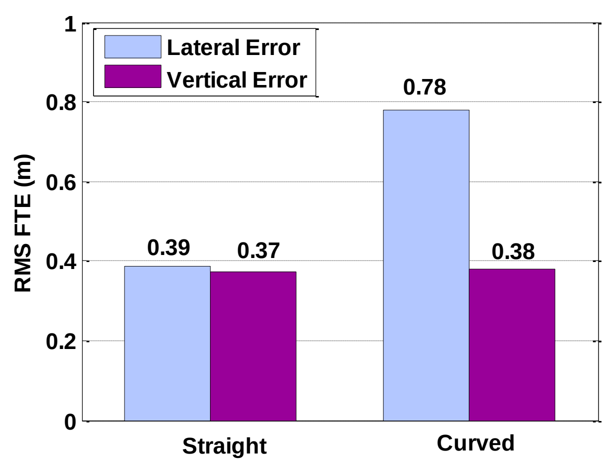

Figure 11. Flight Technical Erorr: Straight vs. Curved Segments

5.3. SEPARATION DISTANCE

From the error models derived above and the *K*-factor determined from the risk assessment in Section 3.5, simulations of safe separation distances between in-network UAVs were conducted. For these simulations, the vertical distance between UAVs was fixed as 20 meters and the horizontal distance between UAVs as 30 meters. We also assumed that the UAVs hover over KAIST campus with both vertical and horizontal velocities of zero. These parameters were applied to compute $\sigma_{tropo,btweenUAVs}$ and $\sigma_{iono,btweenUAVs}$ for NSE. The standard RTCA 24 constellation was used to generate visible satellite geometries. For FTE, the straight flight mode was assumed for both UAVs. Figure 12 shows the result safe separation distances over time as a function of visible satellite geometry, which is known to the UAV network controller. In this figure, the horizontal separation distance, which is shown by the red curve, is mostly below 3 meters. The vertical separation distance varies mostly from 3 to 5 meters reaches a maximum of about 6 meters. These results show what is possible for groups of "in-network" UAVs, but much more work is needed to determine safe separation distances from other vehicles and under different conditions (e.g., under significant winds).

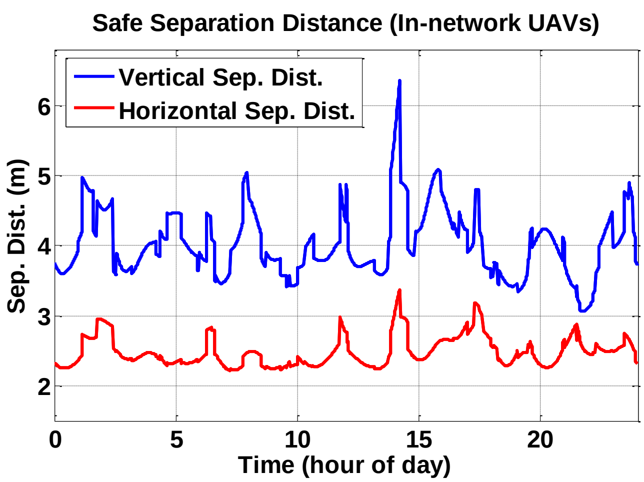

Figure 12. Simulation Results of Safe Separation Distance in the "In-Network" UAVs

6.0. SUMMARY

This paper provides an overview of local area UAV network which optimally guide multiple UAVs and maintain safe separation distance between the UAVs, other vehicles, and the ground. In order to determine safe separation distances, error components which potentially lead to collisions need to be modeled. This paper presents a methodology to model the NSE and FTE both theoretically and experimentally, taking into account the significant correlation of errors among UAVs close to each other.

In this paper, we conducted flight experiments with an octo-copter measuring 110 cm in diameter, 35 cm in height, and 5.31 kg in weight. It is equipped with a GNSS receiver and antenna to collect GNSS measurements along with local-area differential corrections from a nearby reference network. These measurements supported estimation of both multipath (part of NSE) and FTE of the UAV.

From the flight experiments, the lateral FTE RMS on the curved portions showed greater than the vertical error due to the momentum of the UAV when it is taking turns. The results also show that, under the flight conditions tested, the standard airborne multipath error model for manned aircraft also bounds the UAV multipath error with margin. By applying the *K*-factor derived from the safety risk assessment methodology, the proposed error models were used to determine safe separation distances to ensure the safety of UAV operations. In the simulation results for a 24-satellite GPS constellation, horizontal separations between UAVs in the same network are mostly below 3 meters, while vertical separations range between 3 and 5 meters with a maximum exceeding 6 meters.

The separation distances presented in this paper is limited to the case of "in-network" UAVs. Safe separation distances required for different scenarios (e.g., UAVs in the different network, manned aircraft, and ground obstacles) should also be estimated. The NSE and FTE errors highly depend on hardware, environment, and operational conditions and thus need to be evaluated experimentally under various conditions. To support this, future work will include flight tests of several types of UAVs in different flight environments and modes.

ACKNOWLEDGMENTS

The authors thank John Warburton of the FAA William J. Hughes Technical Center and his team for their support. We also would like to thank Oliver Jeannot, Paul Worcester, Dieter Guenter, and Achanta Raghavendra of the AMT Tetra Tech, and Per Enge, Todd Walter, and Sherman Lo of Stanford for their support of this work. Minchan Kim and Kiwan Kim were supported by the KAIST Institute. The opinions expressed here are solely those of the authors.

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